

CYCLAW ADVISOR BRIEFING

LoRA Fine-Tune Kit

What `tools/lora_finetune/` is, what every file does, how it attaches to the live 12-node runtime, and what it is actually worth.

Compiled from two advisor replies on 2026-09-11 against `origin/main`

501d8abcc220a46c4a081675db84eaccfee4c42c (#1368, #1373). Checkout wins over this PDF if main moves.

Direct assessment. This folder is a rented-GPU cookbook that bakes CyClaw house rules into a Qwen 27B overlay. CyClaw the product never imports it. Value is “the local model stops re-deriving I1–I6 every prompt.” It is not a second brain and it is not a knowledge dump of the repo. Landing a trained adapter does not change the six invariants.

1. ELI5

CyClaw already has a local model (`qwen3.8:27b-mlx`) that answers after it **reads your files**. That model does not automatically think like CyClaw. It will happily invent a 13th graph node or suggest wiring writes into `/query`.

LoRA is a small sticker you glue onto a big model. Instead of retraining 27 billion weights, you train a thin overlay on **70** question/answer pairs that say things like: retrieve first, don't let the LLM pick graph edges, don't touch soul without a human reason, don't import `agent` into `gate.py`.

The folder is the sticker factory. You build the 70 examples on any laptop, rent a GPU for an hour, glue the sticker on, export a GGUF, tell Ollama “this is the model now.” Gate, graph, Chroma, soul, audit — unchanged. If you skip the factory, CyClaw still works. RAG is still the encyclopedia; the sticker is the accent.

What 70 stickers cannot do: memorize the repo, replace `config.yaml`, or make a bad retrieval look like a good answer. Overfit is the default failure mode after a few epochs.

2. Tech 101

LoRA / QLoRA in one paragraph

Full fine-tune updates every weight. LoRA freezes the base and learns low-rank matrices ($r=16$ here) on attention + MLP. QLoRA keeps the base in 4-bit so 27B fits in about 24 GB VRAM. CyClaw's kit is QLoRA via Unsloth → TRL SFTTrainer → merged `q4_k_m` GGUF for Ollama.

How it relates to the live stack

<pre>operator CUDA box ----- 70 Q&A → QLoRA on Qwen3.8-27B → adapter + merged GGUF → ollama create cyclaw-qwen</pre>	<pre>CyClaw runtime (unchanged) ----- config.yaml models.local_llm.model: cyclaw-qwen POST /query retrieve → route_by_score → ... local_llm talks to ollama</pre>
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No new graph node. No new OOB package. `verify-deps` E3 skips `tools/`. OSV excludes `r:lora_finetune`. Root requirements / Docker / conda do **not** install Unsloth.

3. Folder and files

Path: `tools/lora_finetune/` on tip 501d8ab. Landed as #1368. #1373 dropped deleted `harness/` from training `source_refs`.

File	Job
<code>curated_qa.py</code>	36 Alpaca pairs: architecture, security, RAG, soul, telemetry, errors, extension, debug, code patterns, philosophy.
<code>cyclaw_debug_qa.py</code>	12 debug scenarios grounded in <code>live_graph.py</code> / <code>INVARIANTS.md</code> / <code>indexer</code> / <code>llm/client.py</code> .
<code>curated_qa_extra.py</code>	22 extras from personality, <code>telemetry_kill</code> , errors, sanitizer, graph.
<code>build_cyclaw_corpus.py</code>	Merges the three tiers into canonical JSON + preview JSONL. Stdlib-first; optional HF tokenizer with SHA-pinned revision.
<code>cyclaw_training.json</code>	The dataset. 70 rows: {id, category, messages[system/user/assistant], source_refs}.
<code>cyclaw_training.jsonl</code>	Preview text column. ChatML fallback if transformers is missing. Train path re-renders with the real Qwen tokenizer.
<code>finetune_qwen38.py</code>	Real Unsloth QLoRA script. CUDA only. Writes adapter + merged GGUF + Modelfile.cyclaw.
<code>dryrun_finetune.py</code>	Fakes Unsloth/TRL/datasets. Proves control flow, 70-row render, <code>max_steps</code> and <code>epochs</code> branches, Modelfile FROM path. No GPU.
<code>requirements.txt</code>	Kit-only pins: <code>unsloth==2026.9.4</code> , <code>transformers==5.17.0</code> , <code>trl==1.13.0</code> , <code>datasets==5.0.1</code> , <code>accelerate==1.15.0</code> .
<code>tests/test_build_corpus.py</code>	Dataset structure, provenance, rendering, JSON round-trip.
<code>tests/test_finetune_integration.py</code>	Mocked control-flow for <code>finetune_qwen38.py</code> .
<code>.gitignore</code>	Ignores <code>_dryrun_out/</code> , rendered JSONL, GGUF artifacts.
<code>README.md</code>	Operator manual: hardware tiers, flags, overfitting guards, model-name caveat.
<code>audit-report.md</code> + <code>findings.*</code> + <code>hardened.diff</code>	Bandit/semgrep pass. Tokenizer from_pretrained pinned to immutable commit SHAs.

File inventory as of origin/main 501d8ab. Draft PR #1380 is kit-test hygiene, not an integration switch.

Dataset shape

70 examples across 10 categories, built from three tiers. Debugging Scenarios = 19 (4 prior + 12 debug + 3 extra). System preamble on every row: reason from source, follow I1–I6, never propose changes that bypass retrieval, audit, or soul governance.

Count	Category
7	Architecture & Topology
8	Security & Defense
5	Code Patterns
19	Debugging Scenarios

Count	Category
4	Design Philosophy
4	Extension Patterns
6	Retrieval & RAG
6	Soul Governance
5	Telemetry & Offline
6	Error Handling

Train loop the kit implements

- Optional regen: `python build_cyclaw_corpus.py`
- No GPU: `pytest tests/ -q` then `python dryrun_finetune.py`
- On ≥ 24 GB NVIDIA: `pip install -r requirements.txt && python finetune_qwen38.py --json cyclaw_training.json`
- Load the **merged GGUF** with `ollama create cyclaw-qwen -f Modelfile.cyclaw` from the gguf dir.

Script flags and defaults (code wins over README)

Flag	Script default	Effect
<code>--model-name</code>	unsloth/Qwen3.8-27B-unsloth-bnb-4bit	HF 4-bit base. Not automatically the same family as shipped qwen3.8:27b-mlx.
<code>--max-seq-length</code>	2048	Every 70-pair example renders under ~ 1500 tokens. Do not raise to 4096 without longer data.
<code>--lora-rank</code> / <code>--lora-alpha</code>	16 / 16	Right size for 70 examples. $r=32/64$ overfits this set.
<code>--max-steps</code>	0	If >0 , caps steps. If 0, uses <code>--epochs</code> .
<code>--epochs</code>	3	Used only when max-steps is 0. ~ 9 steps/epoch at effective batch 8.
<code>--batch-size</code> / <code>--grad-accum</code>	1 / 4	README still says batch 2. Script default is 1. Code wins.
<code>--learning-rate</code>	$2e-4$	adamw_8bit. Drop to $1e-4$ if loss diverges in the first 20 steps.
<code>--no-export-gguf</code>	off (export on)	Skip GGUF + Modelfile; adapter only.
<code>--no-offload-embedding</code>	off (offload on)	Keep the large untied embedding in RAM to save VRAM.

Also on: embedding offload, Unsloth gradient checkpointing, `report_to=none`, PEFT boolean module flags (language / attention / MLP on, vision off) — not a `target_modules` list.

Hardware

- 24 GB (4090 / A10): minimum. Batch 2 or 1 + accum 4 + checkpointing. ~ 1.5 – 2.5 hours for 60 steps.
- 48 GB (A6000 / L40): comfortable. Raise batch, drop accum, cut wall-clock.
- 2×24 GB: Unsloth can DDP; not worth it for 70 examples.
- Apple Silicon M5: **not supported for training**. Unsloth is NVIDIA CUDA. Infer locally; train on RunPod / Vast / Lambda ($\sim \$0.30$ – 0.60 /hr for an A10) or a $2 \times T4$ Kaggle notebook.

4. What 70 examples actually teach

This is a **behavioral / style / debugging-priors seed**, not knowledge injection. 70 examples teach answer shape, invariant discipline (I1–I6), and where to look when a symptom appears. They will **not** teach the full repo. RAG remains the source-of-truth path for code facts. The LoRA nudges reasoning on top of that. Deeper internalization wants on the order of 3–5 million tokens of curated pairs.

Overfit guard. One epoch is ~9 steps at batch 2 / accum 4. Three epochs ≈ 27 steps. The model memorizes exact phrasing fast. Cap --max-steps 60 or --epochs 3. Evaluate on held-out CyClaw questions, not the training set. Rising train accuracy is not progress. If a new symptom does not reach the same diagnostic as current source, add diverse pairs — do not add epochs. An earlier checkpoint (--max-steps 30) often generalizes better than the last one on a set this small.

5. CyClaw integration (almost none, on purpose)

Layer	Talks to the kit?
gate.py / graph.py / mcp	No import. I6 clean. No 13th node. Do not wire safe_generate.
llm/client.py	Indirect only: whatever Ollama tag models.local_llm.model names.
config.yaml	One string you change after ollama create. Shipped default stays qwen3.8:27b-mlx.
Soul / I5	Training text may describe POST /soul/restore and apply_evolution(..., scan=False) on .bak. That is existing escape-hatch documentation inside Q&A, not a new write API.
Spend / Numbat / pre-action hook	Untouched. Train is off-box; no Grok/Claude bill.
Docker / conda / extras / setup-guide	Explicitly out. No install surface pulls this folder.
verify-deps E3 / OSV	tools/ is skipped. OSV uses --experimental-exclude=r:lora_finetune because Unsloth’s published graph still expands to ancient wheels this repo never installs. pip-audit on the GPU box after install.
CI / ruff	Kit tests are optional operator tests. Ruff has a per-file ignore for tools/lora_finetune/tests/*.

The only honest integration is **model identity**. Runtime still speaks OpenAI-compatible to loopback Ollama. assert_local_destination still applies. Graph deadline, max_tokens, and num_ctx still come from config, not from the LoRA. Personality still injects soul.md. Retrieval still builds context. user_gate still triple-gates Grok/Claude. A fine-tune cannot disable those.

Modelfile trap. The script writes `PARAMETER num_ctx 32768` and `temperature 0.3`. CyClaw’s C12 floor is built around shipped `OLLAMA_CONTEXT_LENGTH=16384 + max_context_tokens=8000`. Raising Ollama context to 32k without revisiting C12 / VRAM / stall math can reintroduce the 0% Ollama stall this project already paid to kill. Treat Modelfile `num_ctx` as a suggestion, not a config override.

Safe attach procedure

- Keep the stock tag. Confirm RAG answers a held-out question from retrieved chunks.
- `ollama create cyclaw-qwen -f Modelfile.cyclaw` from the exported GGUF directory.
- Point `models.local_llm.model` at `cyclaw-qwen` on a throwaway config. Do not rewrite the committed default.

- Same held-out question: citations must still come from retrieval. If the model recites a training answer that disagrees with current main, the LoRA is stale. The corpus is a snapshot of 2026-09-07 source, not a live indexer.
- Do **not** slap an adapter onto shipped qwen3.8:27b-mlx unless you have proven the Unsloth bnb-4bit checkpoint is the same weight family. Mismatched bases break LoRA. Merged GGUF is the safe path.

6. What it can be used for

Good uses

- Cut “please remember I2 / don’t add a 13th node / don’t import agentic into gate” slop on local answers.
- Bias debug replies toward route_by_score / user_gate / audit_logger instead of generic LLM folklore.
- Demo: same RAG corpus, two Ollama tags — stock vs cyclaw-qwen — show the house-style delta.
- Personal operator model on the M5: infer locally with MLX/Ollama; train only on a rented NVIDIA card.
- Dataset as documentation: the 70 pairs are a readable invariant catechism even if you never train.

Bad uses — do not sell these

- Replacing the index. 70 examples are not the tree.
- Compliance / air-gap story. A fine-tune can leak training phrasing; it does not add a control.
- Autopilot soul or agentic writes. I5 and the six-gate stay human + source constants.
- Loading an adapter onto qwen3.8:27b-mlx because both names say Qwen3.8-27B.
- Merging Unsloth into the runtime wheel so “it just works.” That explodes OSV, torch pins, and the Apple-silicon install.

7. Value — brutal

Who	Value	Ceiling
You (operator on M5)	Less time restating invariants to the local 27B. Debug priors.	Hours saved per week of heavy local use. Not a product.
CyClaw the codebase	Zero runtime value until you swap the Ollama tag. Isolation value is high: the kit cannot break I1–I6 by existing.	Correct design.
Atlanta MSP / dental-IT pitch	Weak as a line item. Buyers do not pay for “we LoRA’d our own README.” They pay for retrieval that stays on their disk.	Demo spice only.
Job-hunt signal	Shows you know QLoRA and why not to contaminate a runtime lockfile. Easy to overclaim.	One paragraph on a resume, not a product pillar.

Net: the folder is leverage on **local-LLM tone**, not leverage on architecture. RAG + soul + audit remain the product. The LoRA is a nicer local mouthpiece sitting behind local_llm / offline_best_effort.

8. Invariants

I	Verdict if used as designed
I1 RAG-first	Pass only if retrieval still supplies facts. Fail if you treat LoRA memory as canon.
I2 Topology = policy	Pass. No edge changes. No 13th node.
I3 Triple-gate	Pass. Train is local/CUDA; no Grok/Claude bypass.
I4 Audit convergence	Pass. No new audit path. Train is off-box.

I	Verdict if used as designed
I5 Soul governance	Pass if adapter output never calls apply_evolution on its own. Human reason still required.
I6 Module isolation	Pass. tools/ is skipped by dep isolation on purpose. Do not copy kit requirements onto runtime surfaces.

9. Recommendation and checklist

Leave committed `models.local_llm.model` on `qwen3.8:27b-mlx`. Use the kit when you are tired of the stock 27B arguing with `INVARIANTS.md`. Pay for one A10 job, export the merged GGUF, A/B on held-out questions, keep the corpus dated. Do not wait on draft #1380 to “enable” anything — that PR is kit-test hygiene (batch-size / `.gitignore` class), not a runtime switch. Do not promote the LoRA kit into Docker, extras, or OOB.

Check	Bar
Runtime install still has zero Unsloth / TRL / datasets?	Yes on 501d8ab.
Graph still 12-node after ollama create?	Yes. Swap is a model tag only.
Held-out answer still cites retrieved files?	Must. Otherwise I1 got worse.
Base-family verified before any ADAPTER on the MLX tag?	Must. Merged GGUF is the default-safe path.
Modelfile num_ctx reconciled with C12 / 16k stall math?	Must before raising context.
Held-out questions, not training-set recitation?	Required, or you burned GPU money to memorize 70 answers.
Corpus date vs live main?	Snapshot of 2026-09-07 source. Re-build when invariants or deleted packages drift (#1373 already happened).

10. What this is not

- Not a replacement for RAG.
- Not on the request path. Gate never imports it. POST /query still retrieves.
- Not I5. Describing soul restore in Q&A is not a new write path.
- Not a sixth OOB package and not a graph node.
- Not proven by dry-run. `dryrun_finetune.py` does not load 27B weights, does not prove VRAM fit, and does not prove a valid GGUF.
- `audit-report.md` is slightly stale in one sentence (“no requirements.txt in the kit”) even though the file exists now. Trust the files, not that paragraph.

Operator discipline. Until the human has read the actual diff, a fine-tune is just another way to freeze last month’s prose into next month’s answers. Re-list open issues at check time. Do not treat remaining_work.md as current. Do not resurrect harness/ or :8790. Coding/chat console is sibling cgfixit/CG-agent-harness.